

DS 642: Applications of Parallel Computing

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Week 4:

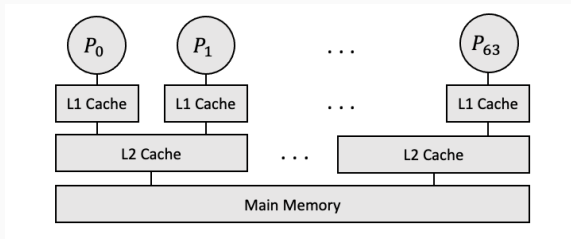
Shared Memory Programming with OpenMP

Parallel Programming Models

Programming model is made up of the languages and libraries that create an abstract view of the machine

- Control
 - How is parallelism created?
- Data
 - What is private vs shared data?
 - How is logically shared data accessed or communicated?
- Synchronization
 - What operations can be used to coordinate parallelism?
 - What are the atomic (indivisible) operations?
- Cost
 - How do we account for the cost of each of the above?

Shared memory vs message passing



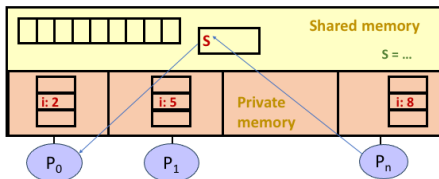
- Implicit communication via memory vs explicit messages
- Threads communicate + synchronize with shared variables

Simple example: Consider the following:

$$\sum_{i=0}^{n-1} f(A[i])$$

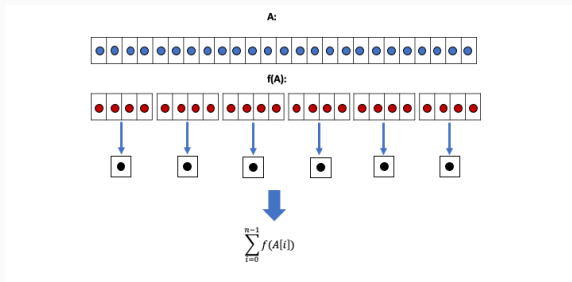
Where does A live? What work will be done by each processor? How the single result is obtained?

Shared Memory Programming Model



- Program is a collection of threads of control.
- Each thread has a set of private variables, e.g., local stack variables.
- Also a set of shared variables, e.g., static variables, shared common blocks, or global heap.
 - Threads communicate implicitly by writing and reading shared variables.
 - Threads coordinate by synchronizing on shared variables

Shared memory strategy



- small number of processors $p \ll n = \text{size}(A)$, attached to a single memory.
- parallel decomposition: each evaluation and partial sum is a task.
- assign n/p numbers to each p .
 - each compute independent private results and partial sums.
 - collect the partial sums and compute global sum.

Data Race Example

Problem is a race condition on variable s in the program:

```
static int s = 0;
```

Processor 1:

```
for i = 0, n/2 - 1;  
    s = s + f(A[i]);
```

Processor 2:

```
for i = n/2, n - 1;  
    s = s + f(A[i]);
```

- A race condition or data race occurs when:
 - two processors (or two threads) access the same variable, and at least one does a write.
 - The accesses are concurrent (not synchronized) so they could happen simultaneously.

Assume $A = [3, 5]$, with $f(x) = x^2$, what do you think that the result of the above code can be?

OpenMP programming model

Program consists of *threads* of control.

- Can be created dynamically
- Each has private variables (e.g. local)
- Each has shared variables (e.g. heap)
- Communication through shared variables
- Coordinate by synchronizing on variables

Introduction to OpenMP

- What is OpenMP?
 - Open specification for Multi-Processing
 - “Standard” application programming interface (API) for defining multi-threaded shared-memory programs
 - openmp.org
- High-level API
 - Preprocessor (compiler) directives (~ 80%)
`#pragma omp construct [clause ...]`
 - Library Calls (~ 19%)
`#include <omp.h>`
 - Environment Variables (~ 1%)

A Programmer's View of OpenMP

- OpenMP is a portable, threaded, shared-memory programming specification with “light” syntax
 - Requires compiler support (C or Fortran)
- OpenMP will:
 - Allow a programmer to separate a program into serial regions and parallel regions, rather than T concurrently-executing threads.
 - Hide stack management.
 - Provide synchronization constructs.
- OpenMP will not:
 - Parallelize automatically
 - Guarantee speedup
 - Provide freedom from data races

Compiling OpenMP Programs

A practical aside...

- Intel has OpenMP support by default

```
icc -c -qopenmp foo.c
```

```
icc -o -qopenmp my_code.exe foo.o
```

- GCC has OpenMP support by default

```
gcc -c -fopenmp foo.c
```

```
gcc -o -fopenmp my_code.exe foo.o
```

- LLVM has OpenMP support by default (more recent)

```
clang -c -fopenmp foo.c
```

```
clang -o -fopenmp my_code.exe foo.o
```

- Apple LLVM *does not* support OpenMP

- And gcc on OS X now aliases clang
- Can use Hombrew for alternate compiler

Parallel “hello world”

```
1  #include <stdio.h>
2  #include <omp.h>
3
4  int main()
5  {
6      #pragma omp parallel // Do this part in parallel
7      printf("Hello world from %d\n",
8             omp_get_thread_num());
9
10     return 0;
11 }
```

Parallel “hello world”

```
1  #include <stdio.h>
2  #include <omp.h>
3
4  int main()
5  {
6      omp_set_num_threads(16);
7
8      #pragma omp parallel // Do this part in parallel
9      printf("Hello world from %d\n",
10             omp_get_thread_num());
11
12      return 0;
13 }
```

The OpenMP Common Core

Most OpenMP programs only use these

OpenMP pragma, function, or clause	Concepts
<code>#pragma omp parallel</code>	Parallel region, teams of threads, structured block, interleaved execution across threads
<code>int omp_get_thread_num()</code> <code>int omp_get_num_threads()</code>	Create threads with a parallel region and split up the work using the number of threads and thread ID
<code>double omp_get_wtime()</code>	Speedup and Amdahl's law. False Sharing and other performance issues
<code>setenv OMP_NUM_THREADS N</code>	Internal control variables. Setting the default number of threads with an environment variable
<code>#pragma omp barrier</code> <code>#pragma omp critical</code>	Synchronization and race conditions. Revisit interleaved execution.
<code>#pragma omp for</code> <code>#pragma omp parallel for</code>	Worksharing, parallel loops, loop carried dependencies
<code>reduction(op:list)</code>	Reductions of values across a team of threads
<code>schedule(dynamic [,chunk])</code> <code>schedule (static [,chunk])</code>	Loop schedules, loop overheads and load balance
<code>private(list)</code> , <code>firstprivate(list)</code> , <code>shared(list)</code>	Data environment
<code>nowait</code>	Disabling implied barriers on workshare constructs, the high cost of barriers, and the flush concept (but not the flush directive)
<code>#pragma omp single</code>	Workshare with a single thread
<code>#pragma omp task</code> <code>#pragma omp taskwait</code>	Tasks including the data environment for tasks.

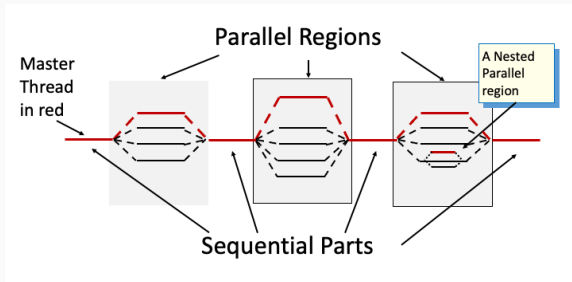
Data scoping: Shared and private

```
#pragma omp construct [clause ...]  
  
1  #include <stdio.h>  
2  #include <omp.h>  
3  
4  int main()  
5  {  
6  ...  
7  
8  #pragma omp parallel private(x)  
9  {  
10 ...  
11 }  
12 }
```

Annotations distinguish between different types of sharing:

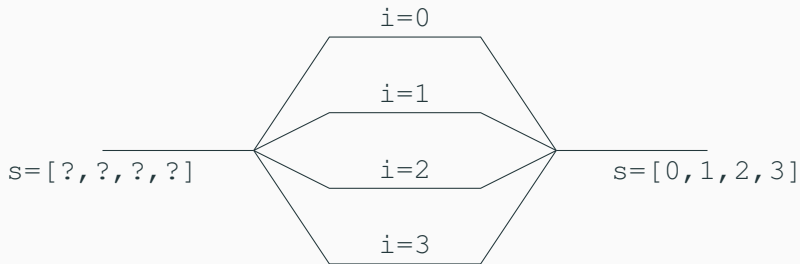
- `shared(x)` (default): One `x` shared everywhere
- `private(x)`: Thread gets own `x` (indep. of master)

Parallel regions



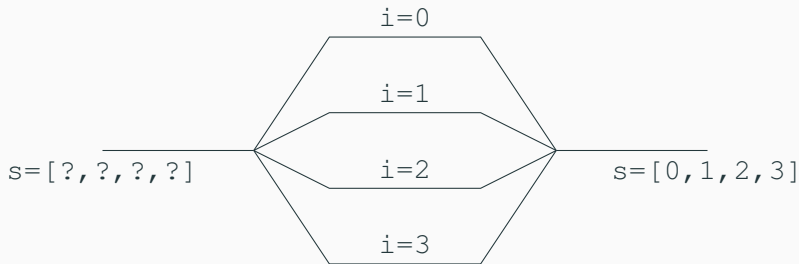
- Basic model: fork-join
- Each thread runs same code block
- Annotations distinguish shared (*s*) and private (*i*) data
- *Relaxed consistency* for shared data

Example: Parallel regions



```
1 double s[MAX_THREADS];
2 int i;
3 #pragma omp parallel shared(s) private(i)
4 {
5     i = omp_get_thread_num();
6     s[i] = i;
7 }
8 // Implicit barrier here
```

Example: Parallel regions



```
1 double s[MAX_THREADS]; // default shared
2 #pragma omp parallel
3 {
4     int i = omp_get_thread_num(); // local, so private
5     s[i] = i;
6 }
7 // Implicit barrier here
```

Several ways to control num threads

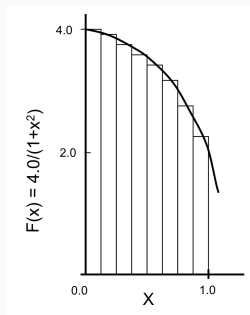
- Default: System chooses
- Environment: `export OMP_NUM_THREADS=4`
- Function call: `omp_set_num_threads(4)`
- Clause: `#pragma omp parallel num_threads(4)`

Can also nest parallel regions.

Parallel “hello world”

```
1  #include <stdio.h>
2  #include <omp.h>
3  int main(int argc, char *argv[])
4  {
5      int nthreads, id;
6      omp_set_num_threads(4);
7
8      #pragma omp parallel private(nthreads, id)
9      {
10         id = omp_get_thread_num(); \\private
11         printf("Hello world from %d\\n", id);
12
13         if (id == 0) {
14             nthreads = omp_get_num_threads();
15             printf("Number of threads = %d\\n", nthreads);
16         }
17     }
18     return 0;
19 }
```

Example: Numerical Integration



$$\int_0^1 \frac{4}{1+x^2} dx = \pi$$

We can approximate the integral as a sum of rectangles:

$$\sum_{i=0}^n F(x_i) \Delta x \approx \pi$$

Serial π program

```
1  #include <stdlib.h>           /* Standard Library */
2  #include <stdio.h>           /* IO library */
3  #include <math.h>            /* Math library */
4  #define pi_true (acos(-1.))  /* double pi_true = 3.14159265358
5
6  int main(int argc, char* argv[]) {
7
8      double h, x, pi=0.; int i; long int n = 100000;
9
10     h = 1. / (double) n; /* Set subinterval size */
11
12     for (i = 0; i < n; i++) { /* Perform integration */
13         x = h * (i + .5);
14         pi += 4.*h/(1. + x*x);
15     }
16     printf(" computed pi = %.16e error = %.16e\n",
17           pi, pi_true - pi);
18     return(0);
19 }
```

Example: Numerical Integration (OpenMP)

Computing π by numerical integration:

```
1  #include <stdio.h>
2  #include <math.h>
3  #include <omp.h>
4
5  #define NUM_THREADS 8
6  static long n = 100000000;
7  double h;
8
9  int main ()
10 {
11     int i, nthreads;
12     double x, pi, sum[NUM_THREADS]={0.};
13     h = 1./(double) n;
14
15     omp_set_num_threads(NUM_THREADS);
16
17     #pragma omp parallel
18     {
19         double x;
20         int id, i, nthrds;
21         id = omp_get_thread_num();
22         nthrds = omp_get_num_threads();
23         if (id == 0) nthreads = nthrds;
24         for (i=id;i<n;i=i+nthrds) {
25             x = h*(i+.5);
26             sum[id] += 4./(1.+x*x);
27         }
28     }
29     for(i=0, pi=0.;i<nthreads; i++)
30         pi += sum[i]*h;
31     printf("\n Computed pi is %.16e with %d threads and error %.16e \n ",pi,nthreads,acos(-1.)-pi);
32     return(0);
33 }
```

OpenMP Summary So Far

OpenMP is a compiler-based technique to create concurrent code from (mostly) serial code

Next time:

- OpenMP can enable (easy) parallelization of loop-based code
- Task-based parallelism
- Other parallel work divisions

More Information:

<https://hpc-tutorials.llnl.gov/openmp/>